

Find and Fix: Parameters, Polar Bounds, and Speed

Answer Key

AP Calculus BC

Quick Answers

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| 1. the corrected $dy/dx = 2.25$, — | 5. the corrected petal area = $\frac{3\pi}{4}$, — |
| 2. the corrected speed = 5.0, — | 6. the corrected distance = 9.073, — |
| 3. the corrected area = 4π , — | 7. the corrected slope = 1.0, — |
| 4. d/dt of $dy/dx = 4t$, the corrected second derivative = 2.0, — | 8. the corrected speed = 8.246, — |
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What is verified

9 of 17 answers machine-checked against SymPy, across 8 problems.
 — marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus BC

FUN-3.C – problems 1, 2, 4, 7, 8

FUN-6 – problems 3, 5, 6

Difficulty 1–5 of 5 across 17 tagged checks.

Common wrong answers

1. If they got 9: reported dy/dt instead of dy/dx
 2. If they got 7: added the two components instead of taking the magnitude
 3. If they got 25.133: integrated from 0 to 2π , which traces the circle twice
 4. If they got 12: differentiated dy/dx with respect to t and stopped there
 5. If they got 14.137: integrated over 0 to 2π , which sweeps every petal twice
 6. If they got 12: integrated the sum of the components instead of the magnitude
 7. If they got -1 : used $dr/d\theta$ as the slope of the tangent line
 8. If they got 6.708: took the magnitude of the position vector instead of the velocity vector
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1. $x = t^2$, $y = t^3 - 3t$. A student reports $\frac{dy}{dx} = 9$ at $t = 2$. (a) Correct it. (b) Name the error.

Solution (a): $\frac{dy}{dt} = 3t^2 - 3$ and $\frac{dx}{dt} = 2t$, so

$$\frac{dy}{dx} = \frac{3t^2 - 3}{2t}, \quad \text{and at } t = 2: \frac{12 - 3}{4}.$$

So the slope is

$$\frac{dy}{dx} = \frac{9}{4} = 2.25$$

Solution (b) — instructor-judged. The student computed $\frac{dy}{dt}$ and stopped. For a parametrised curve the chain rule gives $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy/dt}{dx/dt}$, so the x -derivative must divide in. Full credit gives that reason, not just the formula.

2. $x'(t) = 3t^2$, $y'(t) = 4t$. A student reports speed 7 at $t = 1$. (a) Correct it. (b) Name the error.

Solution (a): At $t = 1$ the velocity vector is $\langle 3, 4 \rangle$, and speed is its length:

$$\sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25}.$$

So the speed is

$$\text{speed} = 5$$

Solution (b) — instructor-judged. Speed is the magnitude of the velocity vector, and the two components point in perpendicular directions, so they combine under a square root rather than by addition. Adding them would be the length of the path only if the motion were along one axis. Full credit names the perpendicularity and the Pythagorean form.

3. Area enclosed by $r = 4 \sin \theta$; a student integrates 0 to 2π and gets 8π . (a) Correct it. (b) Explain.

Solution (a): The correct interval is $0 \leq \theta \leq \pi$:

$$A = \frac{1}{2} \int_0^\pi (4 \sin \theta)^2 d\theta = 8 \int_0^\pi \sin^2 \theta d\theta = 8 \cdot \frac{\pi}{2}.$$

So the area is

$$A = 4\pi \approx 12.566$$

Solution (b) — instructor-judged. $r = 4 \sin \theta$ is a circle of radius 2 centred at $(0, 2)$, and it is drawn completely as θ runs from 0 to π ; continuing to 2π retraces it, so the integral counts every point twice. The check is to find where r returns to zero: $\sin \theta = 0$ at $\theta = 0$ and $\theta = \pi$. Full credit gives that check.

4. $x = t^2$, $y = t^4$, $t > 0$, $\frac{dy}{dx} = 2t^2$. A student reports $\frac{d^2y}{dx^2} = 12$ at $t = 3$. (a) Correct it. (b) Name the omitted step.

Solution (a): First differentiate the slope with respect to t :

$$\frac{d}{dt} \left(\frac{dy}{dx} \right) = 4t$$

Now divide by $\frac{dx}{dt} = 2t$: $\frac{d^2y}{dx^2} = \frac{4t}{2t}$, which at $t = 3$ is $\frac{12}{6}$.

$$\frac{d^2y}{dx^2} = 2$$

Solution (b) — instructor-judged. The second derivative of a parametrised curve is $\frac{d(dy/dx)/dt}{dx/dt}$ — the student produced only the numerator. Full credit states the divided form. (In fact $y = x^2$ here, so $\frac{d^2y}{dx^2} = 2$ everywhere, which is a good independent check.)

5. One petal of $r = 3 \cos(3\theta)$; a student integrates over a full turn and gets $\frac{9\pi}{2}$. (a) Correct it. (b) Explain the limits.

Solution (a): A single petal runs between consecutive zeros of r . Solving $3 \cos(3\theta) = 0$ gives $3\theta = \pm \frac{\pi}{2}$, i.e. $\theta = \pm \frac{\pi}{6}$:

$$A = \frac{1}{2} \int_{-\pi/6}^{\pi/6} 9 \cos^2(3\theta) d\theta = \frac{9}{2} \left[\frac{\theta}{2} + \frac{\sin 6\theta}{12} \right]_{-\pi/6}^{\pi/6} = \frac{9}{2} \cdot \frac{\pi}{6}.$$

So the petal area is

$$A = \frac{3\pi}{4} \approx 2.356$$

Solution (b) — instructor-judged. The limits come from the equation, not from the picture: a petal begins and ends where $r = 0$, so solve $r(\theta) = 0$ and take two consecutive solutions. Integrating over 0 to 2π sweeps all three petals and covers each of them twice. Full credit shows the zeros being solved for.

6. $x'(t) = 2t$, $y'(t) = 3t^2$ on $[0, 2]$. A student adds the two component integrals and gets 12. (a) Correct distance. (b) Name the error.

Solution (a): Distance travelled is the integral of the speed:

$$\int_0^2 \sqrt{(2t)^2 + (3t^2)^2} dt = \int_0^2 t \sqrt{4 + 9t^2} dt = \frac{1}{27} [(4 + 9t^2)^{3/2}]_0^2 = \frac{40\sqrt{40} - 8}{27}.$$

So the distance is

$$\text{distance} \approx 9.073$$

Solution (b) — instructor-judged. Arc length integrates the speed, and speed is a square root of a sum of squares. A square root cannot be distributed across a sum, so integrating each component separately and adding measures neither distance nor displacement. Full credit says exactly that.

7. $r = 1 + \cos \theta$; a student reports the slope at $\theta = \frac{\pi}{2}$ as $\frac{dr}{d\theta} = -1$. (a) Correct slope. (b) Explain.

Solution (a): Convert first: $x = (1 + \cos \theta) \cos \theta$ and $y = (1 + \cos \theta) \sin \theta$. Then

$$\frac{dy}{d\theta} = -\sin^2 \theta + \cos \theta + \cos^2 \theta, \quad \frac{dx}{d\theta} = -\sin \theta (1 + 2 \cos \theta).$$

At $\theta = \frac{\pi}{2}$ these are -1 and -1 , so the slope is their ratio.

$$\left. \frac{dy}{dx} \right|_{\theta=\pi/2} = 1$$

Solution (b) — instructor-judged. $\frac{dr}{d\theta}$ measures how fast the radius grows as the angle turns; the slope of the tangent is a ratio of y - to x -changes in the plane. Getting it requires converting to $x = r \cos \theta$, $y = r \sin \theta$ and applying the product rule to each. Full credit names both product-rule contributions.

8. $x(t) = t^2 - 4t$, $y(t) = \frac{t^3}{3} - t$. A student reports the speed at $t = 3$ as $\sqrt{45}$. (a) Correct speed. (b) Name the error; speeding up or slowing down?

Solution (a): Speed uses the velocity, not the position: $x'(t) = 2t - 4$ and $y'(t) = t^2 - 1$, so at $t = 3$ the velocity is $\langle 2, 8 \rangle$ and

$$\text{speed} = \sqrt{2^2 + 8^2} = \sqrt{4 + 64}.$$

So the speed is

$\text{speed} = \sqrt{68} \approx 8.246$
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Solution (b) — instructor-judged. The student used the position vector; speed is the magnitude of the *velocity* vector, and where the particle is says nothing about how fast it is going. For the second part, $a(t) = \langle 2, 2t \rangle$, so at $t = 3$ the acceleration is $\langle 2, 6 \rangle$ and $v \cdot a = 2(2) + 8(6) = 52 > 0$: the particle is speeding up. Full credit uses the dot-product test (or differentiates the speed) rather than asserting it.